

Cryptography, Key Management & Secure Authentication

1. Objective

The objective of this project is to implement basic cryptographic and secure authentication techniques using Python.

2. Technologies Used

- Python
- AES-256-GCM
- RSA-2048
- HMAC-SHA256
- bcrypt

3. Implementation

- **AES-256-GCM:** Implemented secure encryption and decryption using random IVs.
- **RSA-2048:** Generated public/private key pairs and implemented digital signature verification.
- **HMAC-SHA256:** Implemented message integrity and authentication.
- **bcrypt:** Implemented secure password hashing with salting and work factor.
- **Key Management:** Documented secure storage and key rotation best practices.

4. Testing

All major cryptographic functions were tested successfully, including encryption, decryption, key generation, signature verification, HMAC verification, and password verification.

```
1 from cryptography.exceptions import InvalidSignature
2 from cryptography.hazmat.primitives import hashes
3 from cryptography.hazmat.primitives.asymmetric import padding, rsa
4
5
...
PS C:\Users\aravi\Downloads\cryptography-key-management-secure-auth\cryptography-key-management-secure-auth> python aes/aes_gcm.py
>> python rsa/rsa_signature.py
>> python password/password_hash.py
>> python hmac/hmac_demo.py
Key: 15b776820f0b7ff91c7c3e516110e50af28ebbc7883a3e286cf99e188e5330e8
Nonce: 401c71c213677c04ec8f7003
Ciphertext: a3e08f1cccf23b13fb506b908bc7a66787afaf1c15ce454ce90448beb1e8aef3e18a53e5a9df2759a
Decrypted: This is a secret message.
Round-trip successful: True
Tampered ciphertext rejected: PASS
=== RSA-2048 Digital Signature Demo ===
Signature: a4942f273ee36ea412be12a1ec3b9000f4423c74655a6d233a55686ebd240b983f2b701f81933d6f491d636361fcaec078c109590971af31fbc617e5ce2e26ea5370b9f3e0a0229aaac5
95d2995b31302cbfb2aec9e86126806512255db0983fa2be575658344a8a04f51e875830e82fa5a04f7cba315c8c02f4580b722422a9693f2ba0e10052f23573e172bdc978059a37e82856e67a46ecd
6fa88c17b8ceb041a9d766cf59884e090f98ea2607d3260b77e1ccca7b59242927cada4a19f5e842c0f4f1e9cfa0008948682c3fc73458cd0aa40d0b74d76b661ab3bfddc4822ce3397cb8f558d172b
3add968b36bee7b19578193595fc8df07b6019262ff96a
Original message valid: True
Tampered message valid: False
=== bcrypt Password Hashing Demo ===
Hash: $2b$12$Em31hG0eA,4gg5jLgdmPtulC3oxsA9pE3uL4DvyNr10yax6r3EA6u
Correct password: True
Wrong password: False
=== HMAC-SHA256 Demo ===
HMAC: d2136770071bba7d10de255c3ce6d5702d219ee876b0c56d2d7dfc60a0650f
Original message valid: True
Tampered message valid: False
PS C:\Users\aravi\Downloads\cryptography-key-management-secure-auth\cryptography-key-management-secure-auth>
```

5. Project Status

Status: Completed

All planned implementation tasks have been completed and tested successfully.

6. Result

The project successfully demonstrates secure encryption, authentication, password hashing, and cryptographic key-management practices.

7. Conclusion

This project provided practical knowledge of cryptography and secure authentication and demonstrated how different cryptographic techniques can be used to protect data and user credentials.